

Development of system for streaming environmental metrics data

<https://github.com/jurdabos/vlc>

For the course DLBDSEDE02– tutor: Dr. Sahar Qaadan

STUDENT: TORDA BALÁZS, 92128244 (Data Science Bsc.)

Finalized for submission on 30 Nov 2025

Table of Contents

Abbreviations and definitions	ii
vlc – development phase	1
1. Data system overview	1
2. Development environment and Q for next phase.....	1
3. Reproduce it.....	1
APPENDIX.....	2
Screenshots displaying the milestones during development	2
Diagram illustrating the pipeline.....	4

Abbreviations and definitions

Abbrev	Longform (= translation)
AV	avenida = avenue
c	Celsius
calidad_am	calidad ambiental = air quality
CI/CD	continuous integration/continuous delivery/deployment
CO	carbon monoxide
deg	degree
DLBDSEDE02	Distance Learning Bachelor Data Science Elective Data Engineering – course #2
DR	doctor
DT	Delegación Territorial (regional office of the State Meteorological Agency)
e. g.	exemplī grātiā = for example
etc.	et cetera
fecha_carg	fecha de carga = upload date
FIWARE	Future Internet WARE
geom	types of spatial data that represent geographic features in the context of GIS and spatial DBs
gh	github
JDBC	Java Database Connectivity
m	meter
mm	millimeter
MTBF	mean time between failures
MTTR	mean time to repair
pgJDBC	postgresql Java Database Connectivity Driver
pm	particulate matter
RBAC	role-Based access control
UPV	Universitat Politècnica de València = Polytechnic University of Valencia
yaml	Yet Another Markup Language

vlc – development phase

1. Data system overview

We have designed a data system capable of storing and processing a continuous sensor data stream from the public APIs the municipality of Valencia offers. We have set up two docker networks, one internal and one external. The network security measures include (i) internal network isolation for Kafka/DB/Connect, (ii) single ingress point via reverse proxy, (iii) HTTP Basic auth + thoughtful password design for future RBAC robustness, and (iv) no secrets committed to Git. Two kafka topics have been set up, one for air quality measurements and another one for weather info. CI/CD guidelines have been followed by setting up gh Actions workflow yml. Monitoring has been incorporated to enable sanity checks and MTBF+MTTR calculation for later phases.

2. Development environment and Q for next phase

In the current P2:dev phase, we have implemented and tested the pipeline in an Ubuntu/ext 4 environment on a local machine.

In vlc/infra/, a setup has been sketched and initiated for lightweight VM deployment. A base Azure infrastructure that can be used with an Azure for Students account has been provisioned declaratively via ARM using Bicep. Inside the VM, we use Docker Compose, SQL init scripts and shell automation to configure Kafka, TimescaleDB and Grafana. This combination promises a reproducible infrastructure and application configuration without locking the VLC stack to Azure-specific primitives.

3. Reproduce it

Check the main README.md and follow the instructions to reproduce the pipeline setup.

APPENDIX

Screenshots displaying the milestones during development

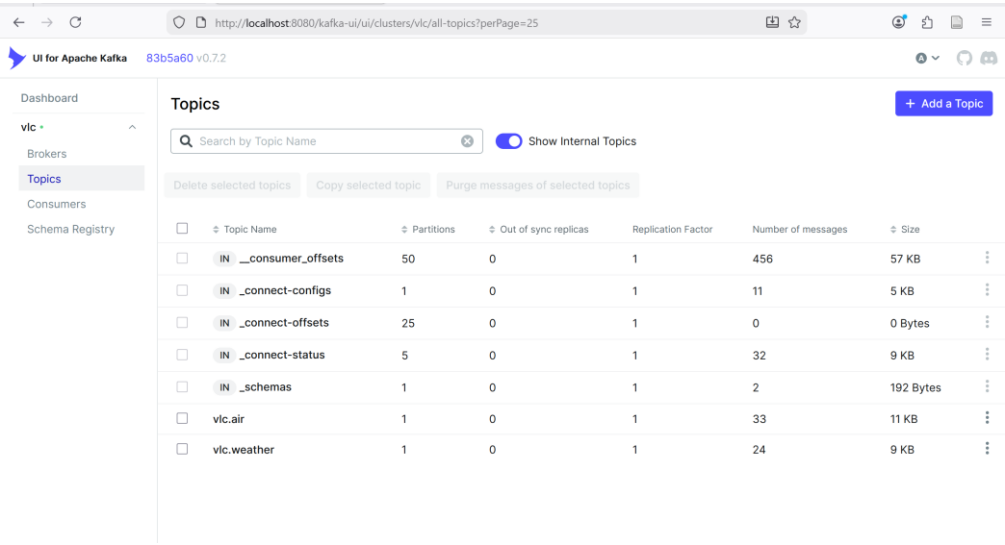


Figure 1: Topics receiving their first pieces of data seen through the kafka-ui

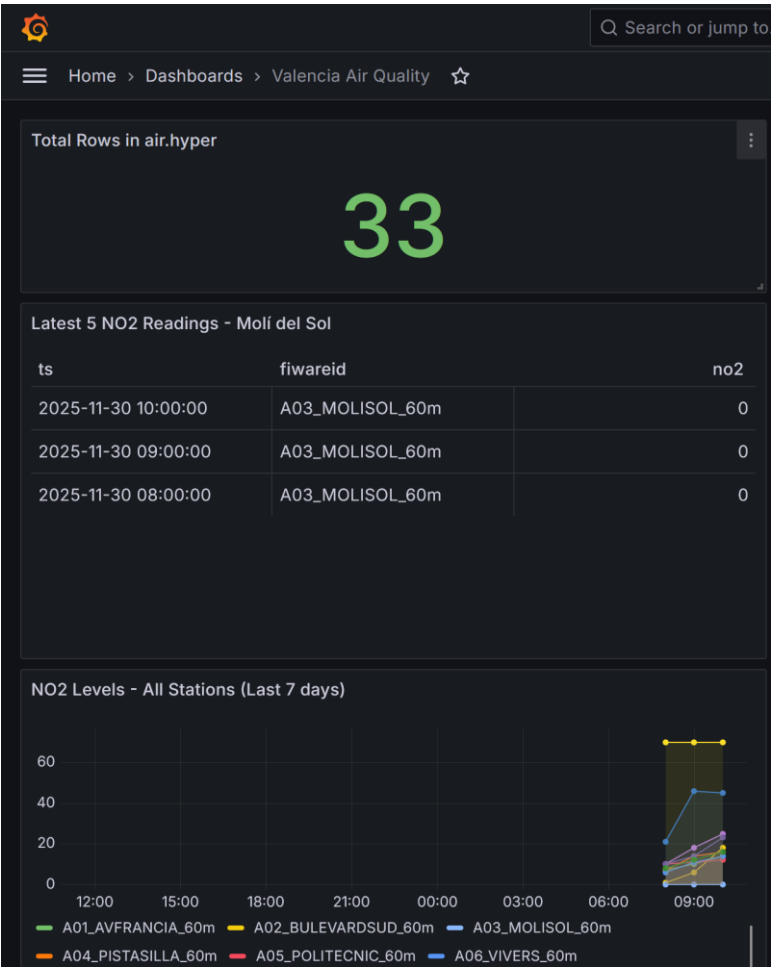


Figure 2: Grafana receiving the first data about rows

fiwareid	records	first_seen	last_seen
A01_AVFRANCIA_60m	15338	2009-01-01	2025-11-30
A02_BULEVARSDUD_60m	14973	2010-01-01	2025-11-30
A03_MOLISOL_60m	15338	2009-01-01	2025-11-30
A04_PISTASILLA_60m	17165	2004-01-01	2025-11-30
A05_POLITECNIC_60m	15704	2008-01-01	2025-11-30
A06_VIVERS_60m	17165	2004-01-01	2025-11-30
A07_VALENCIACENTRE_60m	11763	2018-10-16	2025-11-30
A08_DR_LLUCH_60m	5	2025-11-30	2025-11-30
A09_CABANYAL_60m	5	2025-11-30	2025-11-30
A10_OLIVERETA_60m	5480	2022-01-01	2025-11-30
A11_PATRAIX_60m	5	2025-11-30	2025-11-30
(11 rows)			
fiwareid	records	first_seen	last_seen
W01_AVFRANCIA_10m	5770	2019-01-01	2025-11-30
W02_NAZARET_10m	11043	2020-01-01	2025-11-30
W03_VALENCIAAEROPUERTO_10m	1	2025-11-30	2025-11-30
W04_VALENCIADT_10m	5745	2019-01-01	2025-11-30
W05_VALENCIA_UPV_10m	5046	2021-01-01	2025-11-30
(5 rows)			

Figure 3: Records after a historical backfill run

```
docker exec timescaledb psql -U vlc_dev -d vlc -c "
SELECT
  time_bucket('1 hour', ts) AS hour,
  AVG(temperature_c) AS avg_temp,
  AVG(humidity_pct) AS avg_humidity,
  AVG(pressure_hpa) AS avg_pressure
FROM weather.hyper
WHERE ts > NOW() - INTERVAL '24 hours'
GROUP BY hour
ORDER BY hour;
"
```

hour	avg_temp	avg_humidity	avg_pressure
2025-11-30 06:00:00+00	10.933333333333332	65.33333333333333	1005.2
2025-11-30 07:00:00+00	12.133333333333333	66.83333333333333	1011
2025-11-30 08:00:00+00	12.325000000000001	67.33333333333333	1012
2025-11-30 09:00:00+00	15.554545454545455	57.18181818181818	1013.0909090909091
2025-11-30 10:00:00+00	17.19090909090909	48	1013.3636363636364
2025-11-30 11:00:00+00	17.633333333333336	54.333333333333336	1013
2025-11-30 12:00:00+00	17.983333333333338	55.916666666666664	1013
2025-11-30 13:00:00+00	18.65	52.5	1013
(8 rows)			

Figure 4: Hourly averages across stations since we started up the piping

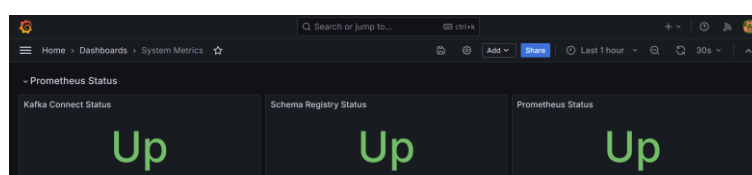


Figure 5: System metrics boasting about services being up

Diagram illustrating the pipeline

